

Harsh Gupta

Software Engineer / Generative AI Engineer

+91-8825334559 | guptaharsh137@gmail.com | [LinkedIn](#) | [Github](#)

Professional Summary

AI / Generative AI Engineer with strong experience in building Agentic systems, RAG pipelines, and scalable backend services using LangChain, FastAPI, vector databases, and cloud-native architectures. Proven ability to optimize latency, cost, and system reliability in production environments.

Education

Jain (Deemed to be University)	June 2023 - August 2025
<i>Master of Computer Application - AI/ML</i>	<i>CGPA 8.0</i>
Lovely Professional University	Apr 2020 - May 2023
<i>Bachelor of Computer Application - Application Development</i>	<i>CGPA 8.55</i>

Technical Skills

Core: Data Structures and Algorithms, Object-Oriented Programming, System Design, Cloud Computing
Generative AI / LLM: LangChain, LangGraph, AI Agents, RAG, Prompt Engineering, Embeddings, Inference, Vector Search, Machine Learning
Languages: Python, C++, C, Java, SQL
Frontend: ReactJS, React Native, NextJS, Streamlit, HTML5, Tailwind CSS, Flutter
Backend: NodeJS, Express, Flask, FastAPI, Dart, Microservices, REST APIs, OAuth, RBAC
Databases: MongoDB, MySQL, DynamoDB, FAISS, Qdrant
Cloud & Tools: Azure, AWS, Git/GitHub, VS Code, Jupyter Notebook

Work Experience

AI Engineering Intern	Aug 2025 - Dec 2025
<i>IQVIA</i>	<i>Bengaluru</i>
- Converted simulation and optimization workflows into agentic systems using LangChain and LangGraph, improving execution reliability and reducing end-to-end workflow execution time from 15 minutes to 2 minutes - Built and benchmarked AI agents across Java, Rust, Go, C#, and JavaScript, evaluating concurrency handling, memory usage, latency, scalability, and power consumption. - Migrated PowerPoint report generation from AWS Lambda to local Node subprocess execution via Python, reducing cost and improving generation latency by 60%. - Built secure, multi-destination execution-data storage across AWS S3, SharePoint, and local environments with OAuth authentication, signed URLs, RBAC, and encryption. - Enhanced overall Promo Suite execution speed by 60% through logic optimization and parallel execution of compute-heavy tasks.	

Software Developer Intern	Jan 2025 - Apr 2025
<i>Ellucian Higher Education Systems</i>	<i>Bengaluru</i>
- Developed reusable cross-platform authentication modules, reducing development time by 90%. - Refined API query logic and response filtering, improving backend performance by 30%. - Integrated Google Workspace APIs with OAuth 2.0 authentication into modular productivity cards. - Proposed and implemented DynamoDB for scalable user data storage, improving retrieval efficiency by 25%. - Authored unit tests using Jest and Playwright to ensure system reliability.	

Software Developer Intern	Apr 2024 - June 2024
<i>Naturemark Systems</i>	<i>Remote</i>
- Contributed to Marketplace systems for Carbon Credit verification, sales, and purchase workflows. - Improved API response quality and reduced payload size by 40% through optimized filtering. - Enhanced image delivery pipelines by implementing image compression, lazy loading, and caching strategies, reducing load time by 20%.	

Projects

School Doubt RAG (Github) <i>NextJS, FastAPI, MongoDB, Qdrant, Ollama</i>	Dec 2025 - Jan 2026
- Designed and deployed a production-grade RAG system processing syllabus PDFs to deliver grade-specific responses with 95% accuracy. - Optimized vector retrieval with Qdrant and custom batching, achieving 100ms latency - reduced inference costs by 60% by running Llama 3 using Ollama on our own server. - Implemented role-based access control for Admins, Teachers, and Students, ensuring secure data isolation. - Architected a comprehensive test suite using Pytest, achieving 90% code coverage across authentication, vector ingestion, and retrieval pipelines.	
Smart Home Automation and Short Circuit Prevention (Patent Filed) <i>React Native, NodeJS</i>	Nov 2024 - Dec 2024
- Developed a smart home system using React Native, Firebase, NodeMCU, voltage sensors, and relays to prevent short circuits and improved safety. - Established real-time communication between IoT devices and backend services with secure authentication. - Reduced electrical hazard incidents through automated fault detection and mitigation.	

Achievements

- Finalist - The Great India Hackathon
- Organizer - India Tour of Open-Source Hackathon, Punjab Edition
- Runner-Up INNOTEK 2023 Innovation Challenge